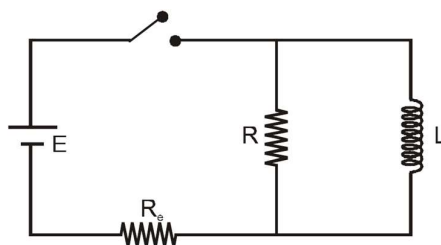
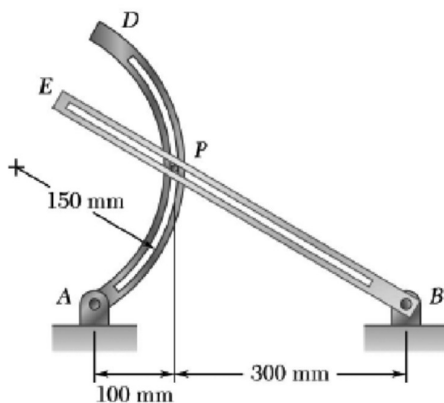


1. In the given circuit Key is closed at $t = 0$. At what time the P.D across inductor is one fourth of emf of the cell.

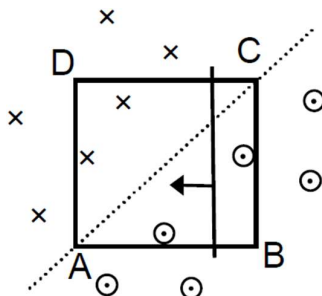


- (A) $\frac{L}{R} \ln^2$ (B) $\frac{L}{R} \ln^4$ (C) 0 (D) $\frac{2L}{R}$
2. The motion of pin P is guided by slots cut in rods AD and BE. Knowing that bar AD has a constant angular velocity of 4 rad/s clockwise and bar BE has an angular velocity of 5 rad/s counterclockwise and is slowing down at a rate of 2 rad/s², determine the velocity of P for the position shown.

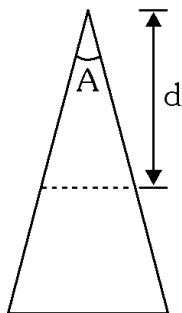


- (A) 0.256 (B) 2.26 m/s (C) 12.26 m/s (D) 22.6 m/s

3. Consider a square frame of side 1m and total resistance 16Ω (uniformly distributed) in two regions of uniform magnetic field. In region left to line AC magnetic field is 200 T inward and perpendicular to the plane. In region right to the line AC magnetic field is 200 T outward and perpendicular to the plane. A conducting rod of negligible resistance slides with uniform velocity 0.5 m/s as shown in the figure. The rod always remain parallel to side BC. Current through the conducting rod when it is 25 cm away from side BC is



- (A) zero (B) $\frac{20}{3}$ A (C) $\frac{40}{3}$ A (D) 10A
4. A very thin prism has an apex angle A and its material has refractive index $\mu = 1.48$. Light is made to fall on one of the refracting faces at near normal incidence. Interference results from light reflected from the outer surface and that emerging after reflection at the inner surface. When violet light of wavelength $\lambda = 400$ nm is used, the first constructive interference band is observed at a distance $d = 3.0$ cm from the apex of the prism.



Choose the **CORRECT** option(s) :

- (A) The apex angle A is 1.6×10^{-4} degree
 (B) If red light ($\lambda = 800$ nm) is used, at what distance from the apex will we observe the first constructive interference band is 6 cm
 (C) The apex angle A is 1.3×10^{-4} degree
 (D) If red light ($\lambda = 800$ nm) is used, at what distance from the apex will we observe the first constructive interference band is 7.5 cm

5. Consider a spherical symmetric distribution of charge. The centre of symmetry is at O. Electric flux linked with a spherical Gaussian surface of radius 'r' and centered at O is given by $\phi = \phi_0 r^4$. Choose the correct option(s).

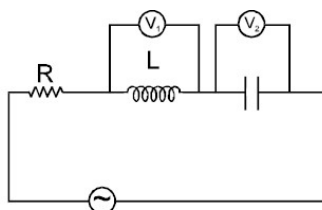
(A) Volume charge density at a distance 'r' from O is given by $\frac{2 \epsilon_0 \phi_0 r}{\pi}$

(B) Volume charge density at a distance 'r' from O is given by $\frac{\epsilon_0 \phi_0 r}{\pi}$

(C) Electrostatic potential energy stored in the spherical region of radius r centred at O is given by $\frac{\epsilon_0 \phi_0^2 r^7}{28\pi}$

(D) Electrostatic potential energy stored in the spherical region of radius r centred at O is given by $\frac{\epsilon_0 \phi_0^2 r^7}{56\pi}$

6. In the circuit shown, resistance $R = 100 \Omega$, inductance $L = \frac{2}{\pi}$ H and capacitance $C = \frac{8}{\pi} \mu\text{F}$ are connected in series with an ac source of 200 volt and frequency 'f'. If the readings of the hot wire voltmeters V_1 and V_2 are same then :



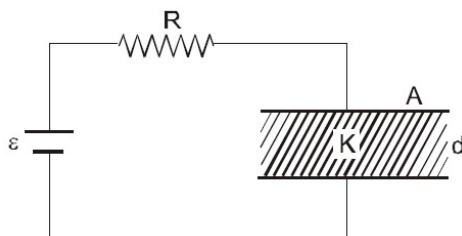
(A) $f = 125 \pi \text{ Hz}$

(B) $f = 250 \pi \text{ Hz}$

(C) current through R is 2A

(D) $V_1 = V_2 = 1000 \text{ volt}$

7. In the circuit diagram a capacitor which is initially uncharged is connected to an ideal cell of emf ϵ through a resistor 'R'. A leaky dielectric fills the space between the plates of dielectric. The capacitance of the capacitor with dielectric is C. Resistance of the dielectric is $R' = R$.



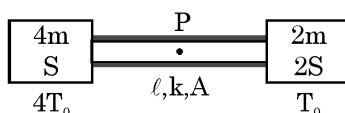
(A) Charge on the capacitor as function of time t is $\frac{\epsilon C}{2} \left[1 - e^{-\frac{2t}{RC}} \right]$

(B) Maximum charge on the capacitor is $\frac{\epsilon C}{2}$.

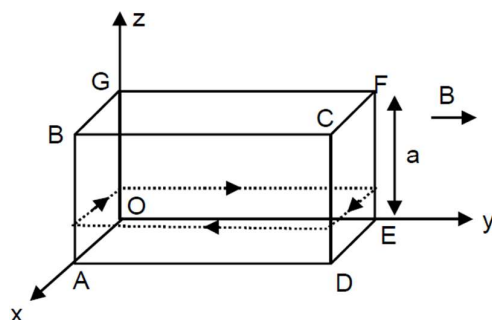
(C) When charge on the capacitor is maximum, then current in the circuit is $\frac{\epsilon}{2R}$.

(D) Maximum voltage across resistor R is ϵ

8. Figure shows two reservoirs containing two liquids of masses $4m$ and $2m$ and their specific heat capacities are S and $2S$ respectively. Their initial temperatures are $4T_0$ and T_0 respectively. The containers are joined by a conducting rod of thermal conductivity K , length ℓ and cross-section area A specific heat capacity of the rod is negligible.

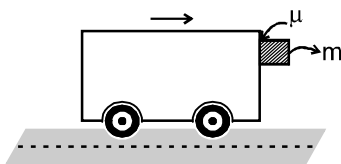


- (A) The temperature of mid-point 'P' of the rod is always constant.
 (B) Heat extracted from container 1 = Heat supplied to container 2.
 (C) After time $t_0 = \frac{2m\ell}{kA} \ln 2$ the temperature difference between containers becomes half of the initial temperature difference.
 (D) After time $t_0 = \frac{ms\ell}{kA} \ln 2$ the temperature difference between containers becomes half of the initial temperature difference.
9. A wooden cubical block ABCDEFGO of mass m and side 'a' is wrapped by a square wire loop of perimeter $4a$ at $z = a/2$ as shown. The current in the wire is I . The whole system is placed on frictionless horizontal surface in a uniform magnetic field $\vec{B} = B_0 \hat{j}$ in this situation. Line of action of normal force between horizontal surface and block passes through a point at a distance x from centre. Select the correct statement(s)



- (A) The block must not topple if $I < \frac{mg}{aB_0}$ (B) The block must not topple if $I < \frac{mg}{2aB_0}$
 (C) $x = \frac{a}{4}$ If $I = \frac{mg}{2aB_0}$ (D) $x = \frac{a}{4}$ If $I = \frac{mg}{4aB_0}$

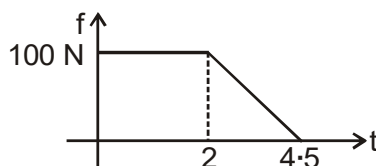
10. The Van is moving on a horizontally ground along a straight road as shown in figure.



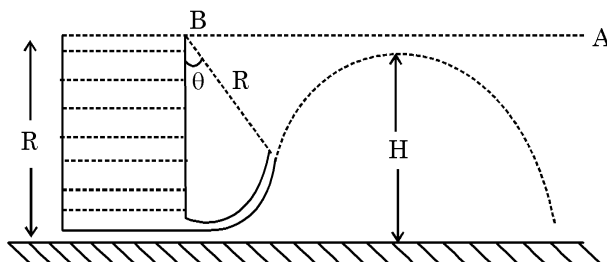
The velocity of Van is given as :

$$v = (9t - t^2 + 10) \text{ m/s, where } t \text{ is in second}$$

Coefficient of friction between the wall of van and block of m kg is μ . The block is at rest w.r.t. van initially. The friction time graph between block and van is given as ($g = 10 \text{ m/s}^2$).



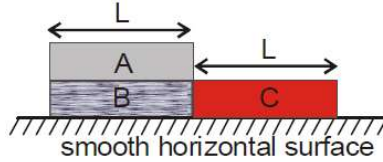
- (A) The mass of block is 10 kg.
 (B) Work done by normal reaction on van is first 5 sec is 202.5 J
 (C) Coefficient of friction between van and block is 0.5.
 (D) Acceleration of block at $t = 3$ sec is 5 m/s^2
11. Figure shows a vessel of depth R having a pipe of small cross sectional area (considered very small as compared to area of vessel) in the shape of a circular arc of radius R as marked. Identify the correct statement.



- (A) $H < R$ for all possible values of θ between 0 to 45° .
 (B) $H > R$ is possible for some angle θ between 45° to 90° .
 (C) Initial value of H for $\theta = 60^\circ$ is more than for $\theta = 45^\circ$.
 (D) $H > R$ is possible for at least 2 different θ between 0 to 90° .

Three identical uniform blocks of mass m each and length L are placed on a smooth fixed horizontal surface as shown. There is friction between A and B (Friction coefficient μ) while there is no friction between A and C.

At the instant shown, that is at $t = 0$; the block A has horizontal velocity of magnitude u towards right, whereas speed of B and C is zero. At the instant block A has covered a distance L relative to block B velocity of all blocks are same.



12. The speed of block A when it just loses contact with B is :

(A) $\frac{u}{2}$ (B) $\frac{u}{3}$ (C) $\frac{u}{4}$ (D) $\frac{2u}{3}$

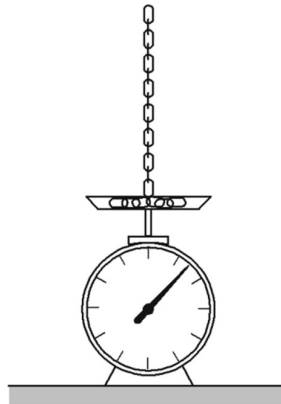
13. The magnitude of total work done by friction on system of three blocks is :

(A) $-\frac{1}{3}mu^2$ (B) $-\frac{1}{4}mu^2$ (C) $-\frac{2}{3}mu^2$ (D) $\frac{1}{3}mu^2$

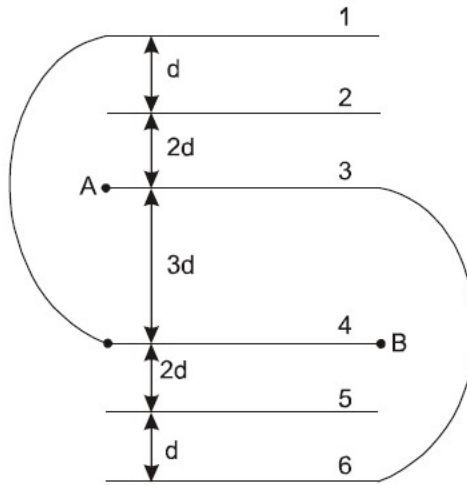
14. The value of μ is

(A) $\mu = \frac{3}{2} \frac{u^2}{gL}$ (B) $\mu = \frac{1}{2} \frac{u^2}{gL}$ (C) $\mu = \frac{u^2}{gL}$ (D) $\mu = \frac{2}{3} \frac{u^2}{gL}$

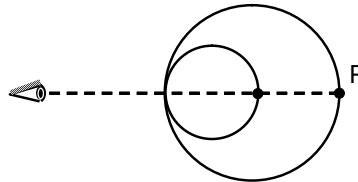
15. A chain of total mass M and length ℓ is suspended vertically with its lowest end touching a scale. The chain is released and falls onto the scale. If is the reading of the scale when a length of chain x , has fallen is $n\lambda gx$. Find n . (Neglect the size of individual links.)



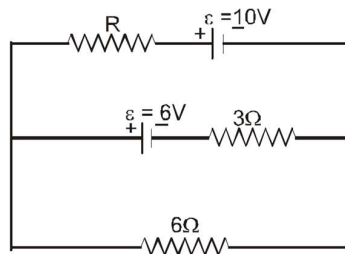
16. There are six plates of equal area A and the plates are arranged as shown in the figure. The equivalent capacitance between A and B is $\frac{N\epsilon_0 A}{d}$, find N .



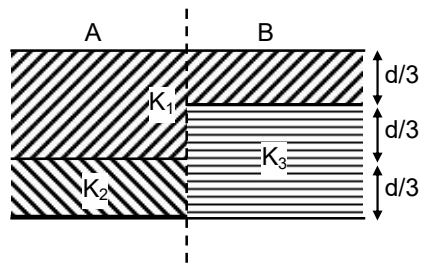
17. A sphere of radius $\frac{R}{2}$ is removed from a solid glass sphere of radius R as shown. Refractive index of glass is 2 and it is surrounded by air. A small bug is sitting at a point P as shown. If the bug appears to a person at a distance of x from the centre of glass sphere (looking from diametrically opposite end) then what will be the value of $\frac{R}{x}$? (Consider only paraxial rays).



18. In the given circuit if the internal resistance of the batteries are negligible, then for what value of resistance R (in Ω) will the thermal power generated in it be maximum.



19. A capacitor is filled with 3 dielectric as shown figure. The value of dielectric constant K_1 , K_2 and K_3 is 2, 4 and 6 respectively. If the ratio of surface charge density on left half to the right half of plate of capacitor to the simplest integer is $\alpha : \beta$ then $\alpha + \beta$ is -



20. In a region, potential energy varies with x as $U(x) = 30 - (x - 5)^2$ Joule, where x is in meters. A particle of mass 0.5 kg is projected from $x = 11$ m towards origin with a velocity ' u '. u is the minimum velocity, so that the particle can reach the origin ($x = 0$). Find the value of u in meter/second.

ANSWER KEY

1. (B)	2. (B)	3. (C)	4. (BC)	5. (BD)
6. (ACD)	7. (ABCD)	8. (ABC)	9. (BD)	10. (AD)
11. (AC)	12. (B)	13. (D)	14. (D)	15. 3
16. 1	17. 18	18. 2	19. 5	20. 12